

Metabolic activity diffusion imaging (MADI): I. Metabolic, cytometric modeling and simulations

Charles S. Springer Jr.^{1,2,3,4,5} | Eric M. Baker¹ | Xin Li^{1,3} | Brendan Moloney¹ | Gregory J. Wilson^{6,7} | Martin M. Pike^{1,4} | Thomas M. Barbara¹ | William D. Rooney^{1,4,8} | Jeffrey H. Maki⁹

¹Advanced Imaging Research Center, Oregon Health & Science University, Portland, Oregon, USA

²Knight Cancer Institute, Oregon Health & Science University, Portland, Oregon, USA

³Brenden-Colson Center for Pancreatic Care, Oregon Health & Science University, Portland, Oregon, USA

⁴Department of Biomedical Engineering, Oregon Health & Science University, Portland, Oregon, USA

⁵Department of Chemical Physiology and Biochemistry, Oregon Health & Science University, Portland, Oregon, USA

⁶Department of Radiology, University of Washington, Seattle, Washington, USA

⁷Bayer Healthcare, Radiology, New Jersey, USA

⁸Department of Behavioral Neuroscience, Oregon Health & Science University, Portland, Oregon, USA

⁹Anschutz Medical Center Department of Radiology, University of Colorado, Aurora, Colorado, USA

Correspondence

Charles S. Springer, Jr., Advanced Imaging Research Center, L452, Oregon Health and Science University, 3181 South West Sam Jackson Park Road, Portland, OR 97239, USA. Email: springer@ohsu.edu

Funding information

OHSU Brenden-Colson Center for Pancreatic Care, Sheppard (PI); Springer (PI); Li (PI) [CSS, XL, EMB, BM]; NIH UL1TR02369, Ellison, Morris (PIs). Awarded by the Oregon Clinical and Translation Institute, Biomedical Innovation Program, Li (PI); Pike (PI) [XL, CSS, EMB, BM]; Bracco Diagnostics, Inc., JHM (PI) [JHM, GJW]

Evidence mounts that the steady-state cellular water efflux (unidirectional) first-order rate constant (k_{io} [s^{-1}]) magnitude reflects the ongoing, cellular metabolic rate of the cytolemmal Na^+ , K^+ -ATPase (NKA), $^{60}MR_{NKA}$ (pmol [ATP consumed by NKA]/s/cell), perhaps biology's most vital enzyme. Optimal 1H_2O MR k_{io} determinations require paramagnetic contrast agents (CAs) in model systems. However, results suggest that the homeostatic metabolic k_{io} biomarker magnitude in vivo is often too large to be reached with allowable or possible CA living tissue distributions. Thus, we seek a noninvasive (CA-free) method to determine k_{io} in vivo. Because membrane water permeability has long been considered important in tissue water diffusion, we turn to the well-known diffusion-weighted MRI (DWI) modality. To analyze the diffusion tensor magnitude, we use a parsimoniously primitive model featuring Monte Carlo simulations of water diffusion in virtual ensembles comprising water-filled and -immersed randomly sized/shaped contracted Voronoi cells. We find this requires two additional, cytometric properties: the mean cell volume (V [μL]) and the cell number density (ρ [cells/ μL]), important biomarkers in their own right. We call this approach metabolic activity diffusion imaging (MADI).

We simulate water molecule displacements and transverse MR signal decays covering the entirety of b-space from pure water ($\rho = V = 0$; k_{io} undefined; diffusion coefficient, D_0) to zero diffusion. The MADI model confirms that, in compartmented spaces with semipermeable boundaries, diffusion cannot be described as Gaussian: the nanoscopic D (D_n) is diffusion time-dependent, a manifestation of the "diffusion dispersion". When the "well-mixed" (steady-state) condition is reached, diffusion becomes limited, mainly by the probabilities of (1) encountering (ρ , V), and (2) permeating (k_{io}) cytoplasmic membranes, and less so by D_n magnitudes. Importantly, for spaces with large area/volume (A/V ; claustrophobia) ratios, this can happen in less than a millisecond. The model matches literature experimental data well, with implications for DWI interpretations.

KEYWORDS

DWI, membrane, Monte Carlo permeability, random walks, stochastic, Voronoi cells, water

1 | INTRODUCTION

1.1 | In vivo $^1\text{H}_2\text{O}$ MR

Water proton ($^1\text{H}_2\text{O}$) magnetic resonance (MR) has numerous in vivo applications. These include noninvasively mapping tissue anatomy/morphology, and even dynamics (e.g., cardiac motion, blood flow). However, $^1\text{H}_2\text{O}$ MR has not been able to report metabolic activity, crucial in all biology. Brain function is normally detected only indirectly, mainly via blood flow changes (functional MRI). Only a handful of nuclear medicine and MR spectroscopic imaging techniques can determine metabolic activities, but with significantly poorer spatial resolution.¹ Addition of metabolic activity capability could have a major impact.

1.2 | The sodium pump; Na^+ , K^+ -ATPase (NKA)

The ubiquitous plasma membrane Na^+ , K^+ -ATPase (NKA) is one of biology's most vital enzymes. This P-type ATPase stores metabolic energy (from ATP hydrolysis) in, and maintains the membrane potential caused by, the respective trans-membrane $[\text{Na}^+]$ and $[\text{K}^+]$ concentration gradients (review¹). The $[\text{Na}^+]$ gradient, in particular, drives many crucial secondary active membrane transporters via Na^+ cotransport. Frustratingly, the very fast homeostatic cellular NKA metabolic rate ($^c\text{MR}_{\text{NKA}}$), which can be 10^{10} $[\text{Na}^+ \text{ and } \text{K}^+]$ ions (cycled)/s/cell, or 90 pmol (ATP consumed by NKA)/s/cell, has never been measurable in vivo.¹ In recent years, however, the active trans-membrane water cycling (AWC) process has been discovered.¹⁻⁷ In AWC, x mole H_2O are cycled across the cell membrane for every mole ATP hydrolyzed by NKA in moving 3 mole Na^+ out and 2 mole K^+ in. The H_2O passes through secondary, active directional water cotransporting membrane symporters, whose rates are thus synchronized to that of the rate-limiting $^c\text{MR}_{\text{NKA}}$. Graphical AWC depictions are shown in fig. 1 of a companion paper⁸ and also in.¹ Brain x values are estimated^{1,8} to be very large, leading to magnitudes such as 10^{12} H_2O molecules actively cycled/s/cell. A list of acronyms and symbols is provided in Table 1.

1.3 | NMR measurement of active trans-membrane water cycling (AWC)

$$k_{io} = k_{io}(p) + k_{io}(a) = \left(\frac{A}{V}\right)P_W(p) + \left(\frac{1}{V}\right)\left(\frac{1}{[\text{H}_2\text{O}_i]}\right)(x)^c\text{MR}_{\text{NKA}} \quad (1)$$

Equation 1 is the first-order differential chemical rate law expression for the homeostatic (steady-state) cellular H_2O efflux (unidirectional) rate constant, k_{io} , where $k_{io}(p)$ and $k_{io}(a)$ are the passive and (metabolically) active k_{io} contributions, (A/V) is the cell surface area/volume (claustraphobia) ratio, $P_W(p)$ is the cell membrane diffusive ("passive") water permeability coefficient, and $[\text{H}_2\text{O}_i]$ is the intracellular water concentration, near 42 M.⁹ (In older literature, the k_{io} reciprocal—the mean intracellular H_2O molecule lifetime, τ_i —is often reported.) Historically, it was believed that the Equation 1 $k_{io}(p)$ term stood alone.¹

It has proven possible to measure k_{io} with the tissue $^1\text{H}_2\text{O}$ MR signal. With current clinical techniques, human MR imaging (MRI) can usually achieve slightly sub-mm spatial resolution. This is significantly better than any other in vivo imaging method capable of estimating enzyme activity and for the first time offers the potential for high-resolution metabolic activity imaging.^{1,5,8} In a recent review,² we tabulate 19 different demonstrations of k_{io} responsiveness to established, investigator-controlled metabolic changes. These independent results clearly demonstrate the presence, often dominance, of the second Equation 1 term. The new k_{io} imaging biomarker could be extremely powerful. Table entries² show potential for the study of cancer, neurological and myocardial pathologies. Because of the central NKA role in intermediary metabolism, k_{io} has diagnostic promise for any tissue or organ.

The work itemized² utilized the longitudinal, trans-cytolemmal NMR shutter-speed (s-s) quantity, $\kappa_{1io} \equiv |R_{1o} - R_{1i}|$, introduced more than 20 years ago¹⁰: R_{1o} and R_{1i} are the intrinsic² longitudinal relaxation rate constants for the extracellular and intracellular $^1\text{H}_2\text{O}$ signals, respectively ($R_1 \equiv 1/T_1$), in the absence of exchange. The s-s concept is inherent to two-site-exchange (2SX) theory² (SI.VII).

1.3.1 | Model systems

The most reliable k_{io} determinations use cell suspensions, superfused viable tissue preparations, and animal models. In some of these, κ_{1io} can be increased sufficiently with large, specified extracellular paramagnetic contrast agent concentrations $[\text{CA}_o]$ or very small magnetic field, B_0 , values

TABLE 1 Acronyms and symbols

A	cell surface area
ADC	apparent diffusion coefficient
ATP	adenosine triphosphate
A/V	cell area/volume [claustrophobia] ratio
AWC	active trans-membrane water cycling
AXR	apparent exchange rate
a_e	membrane “concentration” parameter
$\arg\kappa_{Xio}$	s-s argument
α_i	contracted Voronoi cell i annulus width
α^*	global α target
α_i^*	target α_i
$\alpha_i^{(lim)}$	α_i upper limit
α'	perpendicular distance to gap center
BBB	blood–brain barrier
B_0	main magnetic field
b	acquisition diffusion-encoding potential
CA	contrast agent
CA_o	extracellular CA
$[CA_o]$	CA_o concentration
CG	constant G, t_D incrementation
CI	confidence interval
CT	constant t_D , G incrementation
CXC	constant exchange condition
C_i	standard Voronoi cell i
$C_i^{(cell)}$	inner contracted Voronoi cell i
$C_i^{(ann)}$	annulus of $C_i^{(cell)}$
c_i	seed point of C_i
γ	proton gyromagnetic ratio
D	diffusion coefficient (DT magnitude)
D_A	intrinsic water D in compartment A
D'_A	apparent D_A
D_{ap}	apparent D from Kurtosis Equation
D_B	intrinsic water D in compartment B
D'_B	apparent D_B
DCE	dynamic-contrast-enhanced
DT	diffusion tensor (D)
DWI	diffusion-weighted imaging
D-w	diffusion-weighted
D_{in}	intracellular water D (no exchange)
D'_{in}	apparent D_{in}
D_n	nanoscopic D
D_{out}	extracellular water D (no exchange)
D'_{out}	apparent D_{out}
D_{WM}	D_{in} or D_{out} limit at large t_{RW}
D_0	pure water diffusion coefficient (37°C)
$ \Delta D $	absolute D_{in} , D_{out} difference
d	diffusional water molecule displacement
d^{rms}	root-mean-square of d
ECV	tissue extracellular volume fraction (v_e)
δ	PFG duration
Δ	diffusion interval
Δ_{ap}	apparent Δ
E	expectation
FDG	2-deoxy-2- ¹⁸ Fluoro-D-glucose
FEXI	filter exchange imaging
FOV	field-of-view
FXL	fast-exchange-limit
fL	(1 μm) ³
fMRI	functional MRI
f_M	intracellular macromolecular v
Φ	subset of seeds

(Continues)

TABLE 1 (Continued)

ϕ	spin precessional phase
G	magnetic field gradient tensor
G	G magnitude
GLUT	glucose transporter
GM	cerebral gray matter
Γ	gamma distribution
HPC	high power computing
H_2O_{free}	pure water
H_2O_{in}	intracellular water
H_2O_{out}	extracellular water
$[H_2O_i]$	H_2O_{in} concentration, 0.042 nmol/pL
1H_2O	water proton MR signal
IF	isotropic fractionation
IVIM	intravascular incoherent motion
KA	Kärger/Andrasko
K_{ap}	apparent Kurtosis coefficient
k	steady-state water exchange rate constant
k_{AB}	unidirectional water rate constant, A $\rightarrow$ B
k_{io}	cellular water efflux k (1/ τ_i)
$k_{io(a)}$	active k_{io} contribution
$k_{io(p)}$	passive k_{io} contribution
k_{oi}	cellular water influx k
k_{pe}	capillary CA efflux k
k_{po}	capillary water efflux k
K_o^+	extracellular potassium ion
K-Ras	Kirsten rat sarcoma virus oncogene
κ	fraction of α
κ_{Dio}	diffusigraphic s-s
κ_{Fio}	flow s-s
κ_{Xio}	(X = D, F, 1)
κ_{1io}	longitudinal s-s
Ls-sL	large κ_{Xio} limit (formerly, SXL) condition
Ls-sR	large κ_{Xio} regime (formerly, SXR) condition
l_s	RW step length
l_s^{rms}	root-mean-square l_s
MADI	metabolic activity diffusion imaging
M_{enc}	PFG (zeroth) moment
MR	magnetic resonance
MRI	MR imaging
$^cMR_{enz}$	cellular metabolic rate for enzyme process = $^tMR/\rho$
$^tMR_{enz}$	tissue metabolic rate for enzyme process = $^cMR \cdot \rho$
m	number of membranes crossed in RW step
N	number of samples
NA	normal-appearing
NKA	Na^+ , K^+ -ATPase (sodium pump)
NXL	no-exchange-limit
n	number of seed points
oxphos	oxidative phosphorylation
Ω	domain of Voronoi cells
Ω_{pop}	populated (encompassing) Ω
Ω_{sim}	simulation Ω
Ω_{src}	source Ω
ω	spin precessional (angular) frequency
PET	positron emission tomography
PFG	pulsed G
PVA	partial volume averaging
P(n)	probability of finding n seed points in Ω
p	tissue mole fraction (“population”)
p'	apparent p
p'_A	apparent p of water in compartment A
p'_B	apparent p of water in compartment B
pL	1000 (μm) ³
p_p	water molecule permeation probability

(Continues)

TABLE 1 (Continued)

P_W	membrane water permeability coefficient
$P_W^\dagger$	capillary water permeability coefficient
$P_W(a)$	active cytolemmal P_W contribution
$P_W(p)$	passive cytolemmal P_W contribution
q	PFG wave number (spatial frequency)
RF	radio frequency
ROI	region of interest
RW	random walk
R_1	longitudinal relaxation rate constant
R_{1i}	intrinsic $^1\text{H}_2\text{O}_i$ R_1
R_{1o}	intrinsic $^1\text{H}_2\text{O}_o$ R_1
R_{1oo}	R_{1o} in the absence of CA_o
R_2	transverse relaxation rate constant
r	water molecule position after RW
r_0	starting water molecule position
r_1	longitudinal relaxivity
r_{1o}	CA_o r_1
ρ	cell (number) density
$(\rho V)'$	apparent ρV product
S	tissue $^1\text{H}_2\text{O}$ transverse signal strength
$S(b)$	S at b
S_0	S at $b = 0$
$S^\dagger$	capillary surface area per unit tissue
SD	standard deviation (σ)
SDE	single diffusion encoding
SE	standard error, $\sigma/(\sqrt{N})$
SGLT	sodium/glucose cotransporter
SMS	Siemens Medical Systems
Ss-sR	small κ_{Xio} regime (formerly, FXR) condition
SXL	slow-exchange-limit
SXR	slow-exchange-regime
s-s	trans-cytolemmal MR shutter-speed (κ_{Xio})
T	Tesla
TE	pulse sequence magnetization echo time

TABLE 1 (Continued)

TI	pulse sequence inversion time
TR	pulse sequence repetition time
$\text{Tr}[\mathbf{D}]$	trace of $\mathbf{D}\mathbf{T}$
TT	total acquisition time
T_1	longitudinal relaxation time constant
T_{1-w}	T_1 -weighted
T_2	transverse relaxation time constant
t_D	diffusion time
t_{RW}	RW duration
t_s	RW step time
t_V	t_D when $\kappa_{Dio} = 0$
t_{WM}	t_D when D_n reaches a limiting value
τ_i	mean H_2O_i molecule lifetime ($1/k_{io}$)
V	cell volume
VXC	varying exchange condition
$V(\Omega)$	Ω volume
Vs-s	vanished s-s (formerly, FXL) condition
v	tissue volume fraction
v_b	blood volume fraction
v_e	ECV
v_i	intracellular v (ρV)
WM	well-mixed
x	overall ADC water stoichiometry ($\text{H}_2\text{O}/\text{ATP}$)
x'	x for individual water cotransporter
x	position of random walking molecule
x_k	x after RW step k
$[X_i]$	intracellular X concentration
$[X_o]$	extracellular X concentration
$[X_t]$	tissue X concentration
$\langle X \rangle$	ROI- or voxel-average of X
2SX	two-site-exchange
3S2X	three-site-two-exchange

(near the Earth's field). These selectively increase R_{1o} ($= r_{1o}[\text{CA}_o] + R_{1oo}$; r_{1o} is CA_o relaxivity and R_{1oo} is pre- CA_o R_{1o}), or R_{1i} ,² respectively. CA steady-states can be achieved. Also, complete magnetization recovery can be acquired.²

1.3.2 | Clinical protocols

In clinical studies, however, κ_{1io} cannot be so large: for environmental¹¹ and potential safety¹² reasons, sufficient $[\text{CA}_o]$ cannot be achieved, and (for signal/noise reasons) sufficiently small $\mathbf{B}_0$ diagnostic radiology instruments are not produced commercially. The clinic exists in a s-s “wasteland”.² For human scans, s-s-dynamic-contrast-enhanced (DCE)-MRI pharmacokinetic analysis must be employed, and spatial encoding requires multipulse acquisitions yielding MR signal steady-states.^{1,2} This has its own drawbacks for the determination of k_{io} and the other imaging biomarkers.^{1,2} Nonetheless, many table² entries use s-s-DCE-MRI. The k_{io} trends, if not the magnitudes, so determined, agree with those from the more precise model system determinations. For example, the k_{io} correlation with xenograft melanoma metastatic potential found by s-s-DCE-MRI¹³ is supported by a more reliable determination with xenograft breast cancer cell line suspensions.¹⁴

The model versus clinical distinction has another important basis. To readily measure k_{io} , a s-s law⁸ states that κ_{Xio} must exceed the overall (bidirectional) exchange rate constant k ($\equiv k_{io} + k_{oi}$, where k_{oi} is the H_2O influx (unidirectional) rate constant).² The exchange steady-state condition sets the k_{oi} , k_{io} relationship Equation (2),

$$\kappa_{1io} \geq k = [k_{io} + k_{oi}] = \left[1 + \left(\frac{p_i}{p_o} \right) \right] k_{io}, \quad (2)$$

where p_i and p_o are the sample intracellular and extracellular water mole fractions (“populations”), respectively. Steady-state, CA-bathed cell suspensions or CA-superfused tissues can have small p_i/p_o ratios: k can be almost as small as k_{io} .^{3,4,7} At the same time, very large $[CA_o]$ values can be specified and achieved: thus k_{1io} can easily exceed k , and k_{io} be determined with precision. This is true even although isolated breast cancer cells have been reported to have k_{io} values ranging from 7 to 77 s^{-1} (depending on strain and metabolic condition),^{14,15} or 19 s^{-1} for glioma¹⁶; the latter employed a large diffusigraphic s - s (k_{Dio} ,¹⁷ see below). For $k_{io} = 40 s^{-1}$, $[CA_o]$ near 40 mM is required,² and this can be attained in cell suspensions.^{14,15}

However, the in vivo p_i/p_o ratio is large, often near four. For cells with $k_{io} = 40 s^{-1} cell^{-1}$, k could be $5k_{io}$, $\sim 200 s^{-1}$. Large cell k_{io} values can be partial volume “diluted” in $\langle k_{io} \rangle_{ROI}$ measurements spanning entire tumor or tissue regions of interest (ROIs),^{1,6,14,18} possibly reducing $[CA_o]$ required. But pixel-wise parametric mapping is a different matter. For example, in vivo normal prostate parenchymal k_{io} values may be too large for ss-DCE-MRI k_{io} maps,¹⁹ generally limited to k_{io} less than 10 s^{-1} . After the clinical DCE-MRI bolus injection, the $[CA_o]_{max}$ value can hardly reach 3 mM,²⁰ less than 1/10 required, and it is very transient (a few minutes at most). Thus, although s - s -DCE-MRI k_{io} trends seem reliable, absolute values are questionable: k_{io} may be considerably underestimated. Some assume in vivo k_{io} is always inaccessible.²¹ Furthermore, for the in vivo brain, CAs do not cross the intact blood–brain barrier (BBB). $[CA_o]$ is negligible.

There are other important points: although k_{io} and k_{oi} are both pseudo-first-order rate constants, k_{io} is the less confounded: it is formally independent of the tissue cell (number) density, ρ (cells/ μL [tissue]), while k_{oi} varies with ρ . Equation 1 shows that k_{io} depends only on cell-based properties ($[A/V]$, $P_W(p)$, $\langle V \rangle$, x , cMR_{NKA}). However, Equation 2 shows that k_{oi} also depends on cell density ρ : $p_i = (\rho \bullet \langle V \rangle + f_M)/(1 - f_M)$, where f_M is the intracellular macromolecular volume fraction. We have characterized the k_{io} parameter as supra-intensive.^{1,5} This is a property it shares with the nuclear decay rate constant (the “half-life” basis of radio-dating). Any s - s method also yields p_i , which is proportional to v_i , the intracellular volume fraction.²² But $v_i = \rho \bullet \langle V \rangle$, where $\langle V \rangle$ is the mean cell volume in the ROI or voxel. The $\rho \bullet \langle V \rangle$ product cannot be separated by a s - s approach (Equations S16–S20; see the supporting information). We find the cytometric ρ and $\langle V \rangle$ parameters to be important biomarkers in their own right, and also have potential for comparison with independent estimation from ex vivo histopathology.⁸

For these, and the environmental/safety reasons, we have sought a truly noninvasive, s - s -free method. For early pathology detection and frequent therapy monitoring, one does not want repeated CA injections. The shutter-speed approach has allowed the discovery of active water cycling, and given a glimpse of k_{io} richness,² but moving past s - s is imperative for pursuing in vivo k_{io} further.

1.4 | Diffusion-weighted MRI (DWI)

One of the largest noninvasive MRI subfields employs water diffusion-weighted 1H_2O signals (DWI). There are over 3000 publications/year covering essentially every tissue and pathology (clinical review²³). In preliminary studies, we found that parsimonious Monte Carlo random walk (RW) simulations of water diffusion in digital 3D ensembles of identical spheres matched experimental data well, in gratifying consistency with estimates from independent parameter measurements (including k_{io}), even with no formal “fitting”.¹ These determine k_{io} with no complication from k_{oi} , and also yield ρ and $\langle V \rangle$ independently. However, the compartmental monodispersity caused a “diffraction” artifact (nonrandom residuals), and with identical spheres v_i cannot exceed 0.74. We sought a parsimonious, polydisperse approach that can reach past the common biological tissue v_i of 0.8, and report herein a next generation effort, which we call metabolic activity diffusion imaging (MADI).

2 | METHODS

Computational aspects of our contracted Voronoi cell ensemble creation, and generation of the RW-simulated displacements and DWI decays, are detailed in the supporting information.

The simulation software was written in *Python* and leverages the popular numerical packages *Numpy*²⁴ and *Scipy*.²⁵ These tend to wrap routines written in *C/C++*, or *Fortran*, to provide high performance. This approach allows for both high development speed and high runtime performance.

Specifically, the *Scipy* package provides routines for creating contracted Voronoi tessellations, and for computing polyhedral surface areas and volumes. It also provides a kD-Tree data structure that allows efficient k nearest neighbor queries. Before each simulation starts, the cell “centers” are used to construct such a kD-Tree.

We perform the simulation for batches of 10,000 water molecules at a time, with all molecular positions stored in a single *Numpy* array. The position array is randomly initialized and then used to perform an initial query against the kD-Tree to identify the two nearest neighbor cells for each molecule. This allows us to locate each molecule as either inside a cell or in the interstitial space, and label it with the relevant cell indices. On each iteration, we add an array of randomized displacements to produce a new “proposed” position array and repeat the kD-Tree query on these new positions. Comparing old and new labels for each molecule allows us to determine which ones must pass through one or more cell

membranes. A uniform random variable is sampled for each such molecule to determine if the passage succeeds, with the position being reverted to the old one for those that did not.

This batching approach allows us to maximize the amount of time spent in the fast *C* and *Fortran* routines versus in the slower *Python* routines. Using this approach, we achieve 330,000–375,000 steps per second per CPU core (using a 2.4 GHz Intel E5-2630L), depending on the cell density. With a simulation of 80,000 steps per walk, and the necessity for roughly 12×10^6 walks per library entry, we need typically ~ 15 CPU hours per library entry (this increases with ρ). This many RWs are necessary to avoid simulation artifacts (nonrandom residuals). Because Monte Carlo simulation is an exceedingly parallel process, we can run each simulation in parallel. Therefore, we simulate many different library entries in parallel. Our software package includes support for automatically leveraging both of these avenues for parallelism on a high-performance computing (HPC) cluster, OHSU ExaCloud in this case.

3 | RESULTS

3.1 | Digital cell ensembles

Figure 1 is a 3D rendering of a digital ensemble of randomly sized/shaped, contracted “Voronoi” cells; stochastic geometry.²⁶ This ensemble has $\rho = 153,000$ cells/ μL (the typical $^1\text{H}_2\text{O}$ MRI voxel volume), $\langle V \rangle = 5.6$ pL (1 pL = $1000 [\mu\text{m}]^3$), $v_i = 0.72$, and interstitial gaps of 4 μm or less. These are feasible tissue values. The portion shown (with edges measured in μm) occupies $\sim 1/8$ μL , and has $\sim 19,000$ cells.

Ensemble creation begins by stipulating a v_i value. As above, a given v_i corresponds to an infinite number of possible $\rho\langle V \rangle$ pairs. The resulting $\rho\langle V \rangle$ plane hyperbola can be discretized. In this way, a few v_i values efficiently yield many different ρ and $\langle V \rangle$ values that can be independently searched (although pairs on only extant hyperbolae—some are shown in Figure S3c—are available). We then select a discrete ρ value (for this v_i), thus determining its corresponding $\langle V \rangle$ value. The specified ρ value is “realized” with a random Poisson spatial distribution of cell “centers”.

A standard Voronoi cell ensemble has no interstitium, which we require. Thus, from no extracellular space (i.e., $v_i = 1$; $\langle V \rangle_{\text{max}} = 1/\rho$), we shrink the cells almost uniformly. Each imaginary interstitial cell annulus grows to less than a target thickness, α^* , or a given fraction of the center-to-center distance. A graphic elaboration of this is given in Figures S1 and S2; the procedure is detailed in the supporting information. Figure S3 shows the relationship between v_i and α^* .

Despite its complicated appearance, Figure 1 enables primitively simple (parsimonious) modeling that, in its basic attributes: (1) is avascular, (2) is macroscopically isotropic, (3) uses the pure water (unrestricted) diffusion coefficient [D_0] for the nanoscopic RW steps (the nanoscopic diffusion coefficient, D_n), (4) ignores cell morphology, and (5) does not have a shutter-speed. Attribute 1 is possible because vascular DWI effects are usually small, or can be experimentally nullified.²⁷ Number 2 means we neglect the anisotropic nature of the diffusion-weighted (D-w) signal. Anisotropic DWI is the goal of a large percentage of elegant studies (fundamental reviews^{28,29}). Our approach is complementary to these tissue microstructure studies: we focus on the **D** tensor magnitude, D , not its anisotropy. Attribute 3 reflects that an overall D is not a single-valued quantity for tissue (see below). Here, every compartment, and the interstitium, is filled with pure water. Number 4 constitutes a spatial ergodic approximation: there is intravoxel partial volume averaging (PVA). As implied above, D-w $^1\text{H}_2\text{O}$ signals are “colligative”, in the sense they are averaged over at least tens of thousands of cells, of different types, sizes, and shapes.

Our basic hypothesis is that tissue water DWI signatures (“fingerprints”) can be absolutely, quantitatively described with three fundamental, cytometric, and metabolic activity properties: (1) ρ , the cell number density, (2) $\langle V \rangle$, the voxel- or ROI-average cell volume, and (3) the mean cell, $\langle k_{io} \rangle$. (It is important to note that while k_{io} reports steady-state water exchange kinetics, a $\langle V \rangle$ change reports net trans-membrane water transport.) We use no empirical (arbitrary) variables; which could not, in principle, be independently determined. As described above, the ρ value is specified at each ensemble creation. Then, a literature numerical method determines the $\langle V \rangle$ value and the mean cell surface area, $\langle A \rangle$, value for each ensemble via a separate Monte Carlo process sampling 5×10^6 random cell shapes expected in a given multicell ensemble, for each of 26 different α^* (thus, v_i) values (elaborated in the supporting information). With care, this approach can evaluate multicellular ensembles with any allowed $\rho\langle V \rangle$ pair (Figure S3c). Therefore, the ρ and $\langle V \rangle$ parameters are manifest in the ensembles themselves.

3.2 | Monte Carlo random walks

The k_{io} parameter is manifest in ensemble RWs, because its value is input for each. Typically, 10^5 RWs are conducted in each of 120 randomly different ensembles with the same $\langle k_{io} \rangle$, ρ , and $\langle V \rangle$ parameters, to average over a large number (12×10^6) of RWs. Figure 2 is a 2D cartoon depicting a Monte Carlo RW in an ensemble portion. It starts at position r_0 . Steps inside and outside cells are light and dark blue, respectively. At the end of the RW period, the water molecule had reached position r : the net displacement vector, $\mathbf{d}$, is colored red. The pulsed magnetic field gradient (PFG) vector ($\mathbf{G}$) will be required below. There have been many numerical (Monte Carlo, finite difference) simulations of tissue water diffusion (some recent reports^{29–39}); even one with Voronoi cells.³⁷ Cell membrane water permeation has been considered important since essentially the

beginning of DWI.⁴⁰ Usually, this has been investigated in terms of the water permeability coefficient (P_W) and/or the permeation probability, p_p , the fraction of H_2O molecule/membrane encounters resulting in a permeation.³⁰ Although Equation 1 makes the passive $P_W(p)$ explicit, it implies an active contribution, $P_W(a)$, as well. This is elaborated below. A Monte Carlo RW analysis implicitly evaluates $P_W(p)$.

For Monte Carlo evaluation of the first-order chemical trans-membrane transport rate law, two important relationships have been derived: (1) $P_W(p)$ is proportional to p_p (Equation 3),

$$P_W(p) = p_p \left(\frac{D_0}{l_s} \right) = p_p \left(\frac{D_0}{\delta t_s} \right)^{1/2}, \quad (3)$$

where l_s and t_s are the RW step length and step time, respectively,⁴¹ and $D_n = D_0$; and (2) it is the cell claustrophobia ratio, (A/V) , that relates $P_W(p)$ (immeasurable in vivo) to the cellular $k_{io}(p)$ (Equation 4).⁴²

$$\langle k_{io}(p) \rangle = \left\langle \left(\frac{A}{V} \right) P_W(p) \right\rangle. \quad (4)$$

Equation 4 embodies the supra-intensive nature of steady-state $k_{io}(p)$ (its p -independence): $\langle \rangle$ represents the ensemble average. Combining Equations 3 and 4, Equation 5 allows determination of $\langle k_{io}(p) \rangle$ for our RWs.

$$\langle k_{io}(p) \rangle = \left\langle \left(\frac{A}{V} \right) p_p \left(\frac{D_0}{\delta t_s} \right)^{1/2} \right\rangle = p_p \left(\frac{D_0}{\delta t_s} \right)^{1/2} \left\langle \frac{A}{V} \right\rangle \quad (5)$$

We evaluate $\langle A/V \rangle$ for each ensemble (see the supporting information), we fix D_0 ($3.0 \mu m^2/ms$; at $37^\circ C$) and t_s ($1 \mu s$: therefore, $t_{RW,max} = 80 ms = 80,000$ RW steps), and vary p_p (representative value; 0.005) to vary $\langle k_{io} \rangle$ for different RWs. [$\langle l_s \rangle = (12 D_0 t_s / \pi)^{1/2} = 134 nm$.]

Simulating k_{io} in this way obviously means we are treating it as only passive: the Equation 1 first term, $k_{io}(p)$. But experimental data include the second, active term, $k_{io}(a)$. This can be understood as an active contribution to P_W : $P_W = P_W(p) + P_W(a)$. From Equation 1, we obtain $P_W(a) \approx \{1/([H_2O_i] \langle A \rangle)\} x^c MR_{NKA}$ (it is only approximate because $\langle A/V \rangle \approx \langle A \rangle / \langle V \rangle$, i.e., not identical, but similar; Figure S7). Thus, active water cycling reflects NKA enzyme activity affecting homeostatic membrane water permeability. If one likes to envision cell membrane permeation in terms of “pores”, these pores have transient openings and closings controlled by metabolic activity. The p_p and $x^c MR_{NKA}$ factors bear some analogy.

An important property of true parallel process rate constants is their additivity.¹ Thus, $k_{io} = k_{io}(p) + k_{io}(a)$. Although we simulate as if $k_{io}(a)$ is zero, the k_{io} value from experimental data best-matched by a simulated curve will contain the very informative $k_{io}(a)$. This is a crucial consideration.

3.3 | Diffusional displacements

First, we report the (RW time) t_{RW} -dependences of root-mean-square molecular displacements ($d^{rms} = (d^{ms})^{1/2} = \langle d^2 \rangle^{1/2}$) for 10^6 RWs with no D-w MR encoding (i.e., $G = 0$) in randomly different ensembles, each with $\rho = 781,000$ cells/ μL and $\langle V \rangle = 1.0 pL$ ($v_i = 0.78$). If we set $\langle k_{io}(p) \rangle = 0$ (i.e., $p_p = 0$; Equation 5), we can observe separate displacement time-courses for H_2O_{in} and H_2O_{out} (Figure 3a). The Einstein equation for a 3D Gaussian diffusion process in a homogeneous fluid is: $d^{ms} = 6 D_0 t_{RW}$ ($t_{RW,max} = 80 ms$, here). Thus, for such a case (pure water), the $d^{ms} = f(t_{RW})$ plot is linear (black dashed line for H_2O_{free} ; Figure 3a) with slope $6 D_0$: note the squared-scaled Figure 3a ordinate. However, as expected, the H_2O_{in} (blue) and the H_2O_{out} plots (orange) are nonlinear—at their very origins—but the H_2O_{in} trace eventually becomes horizontal.⁴³

To emphasize the last sentence, Figure 3b,e graphs the t_{RW} -dependences of the Figure 3a instantaneous slope/6 values. Thus the plots are labeled D_0 (for free water), D_{out} , and D_{in} . The free water plot is a horizontal dashed line at D_0 ($3 (\mu m)^2/ms$). However, for the Figure 3a curves, the slopes are t_{RW} -dependent: they start ($t_{RW} = 0$) at D_0 but, within 50 ns, begin to decrease. The D_{out} curve decreases faster, reaching a limiting value, D_{WM} , near $1.8 \mu m^2/ms$, while the D_{in} curve reaches $D_{WM} = 0 (\mu m)^2/ms$ more slowly (after $\sim 20 ms$). After D_{WM} is reached, compartmental water can be considered “well-mixed”: diffusion has entered the homeostatic, permeation-limited regime (elaborated in the Discussion). There are no analytical expressions for the Figure 3a,b,e curve shapes. Diffusion confined within a single sphere was analyzed years ago⁴⁴: the D_{in} limit is zero. For that case, the well-mixed condition is approached when d^{rms} equals the sphere radius⁷; (Equation 6),

$$t_{RW} (well\ mixed) > \frac{V^{2/3}}{17 D_0}, \quad (6)$$

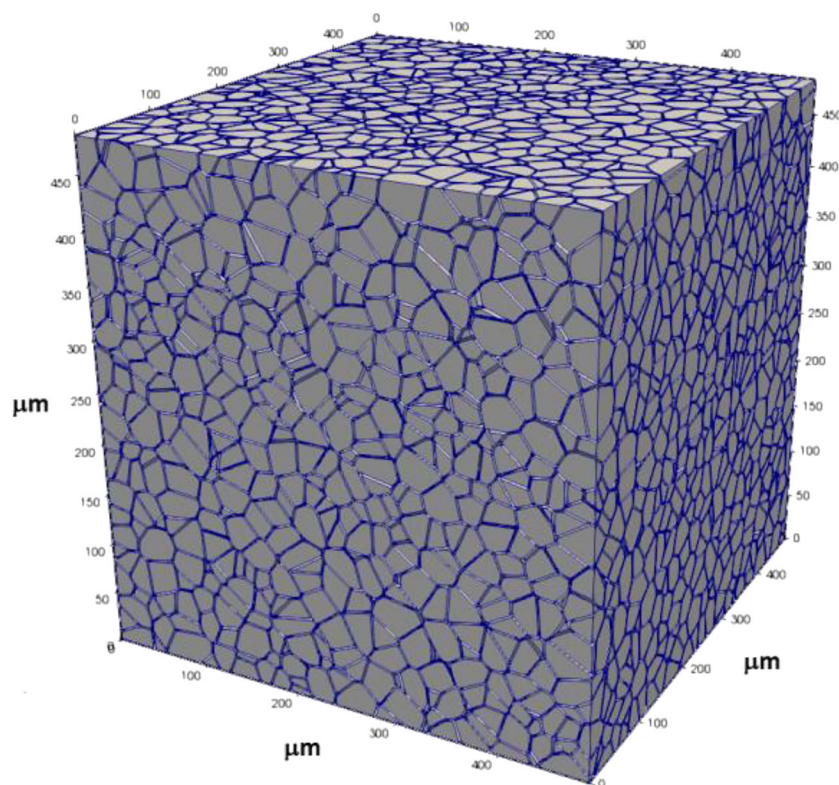


FIGURE 1 Virtual tissue. A virtual ensemble of contracted Voronoi cells ($[500 \mu\text{m}]^3$; $\sim 1/8$ $^1\text{H}_2\text{O}$ MRI voxel), $\sim 19,000$ cells occupying 72% of space ($v_i = 0.72$), mean cell volume ($\langle V \rangle$) 5.6 pL, and cell number density (ρ) 153,000 cells/ μL

with V in $(\mu\text{m})^3$ (10^{-3} pL) and D_0 in $(\mu\text{m})^2/\text{ms}$. For a single sphere with $V = 1000 (\mu\text{m})^3$ (1.0 pL) and $D_0 = 3 (\mu\text{m})^2/\text{ms}$, t_{RW} (well-mixed) > 2 ms. The Figure 3b,e D_{in} curve, for an ensemble with $\langle V \rangle = 1.0$ pL, exhibits such behavior. For $V = 0.2$ pL, t_{RW} (well-mixed) ≥ 0.7 ms. The D_{out} time-course surely varies with ρ as well as $\langle V \rangle$ (more generally, $\langle A/V \rangle$).

At these tissue-mimetic ρ and $\langle V \rangle$ values, the extracellular water molecules encounter diffusional hindrance (cell membranes) much sooner than the intracellular molecules do (Figures 2 and 3b,e). Importantly, they become well mixed more quickly because of the larger extracellular $\langle A/V \rangle$ value; visualized in Figure 2 (dark blue steps).

Even though each H_2O molecule has $D_n = D_0$ for the ensemble random walks, an overall D value is not time-independent: neither ensemble displacement time-course reflects Gaussian diffusion. For a compartmentalized system, the concept of a single-valued overall D does not apply: many use the deliberately vague term “diffusivity”, or sometimes “mobility”. Just the presence of impermeable or semipermeable cell membranes imparts infinitely non-Gaussian character.

As indicated in Figure 3b, the absolute difference between $D_{\text{out}}(t_{\text{RW}})$ and $D_{\text{in}}(t_{\text{RW}})$ is equal to κ_{Dio}/q^2 , where κ_{Dio} is the “diffusigraphic, trans-cytolemmal NMR shutter-speed”¹⁷ and q is the PFG “wave number” (spatial frequency, see below), when there is a PFG. In D-w MR, 2SX theory is manifest as the Kärger/Andrasko model (SI.VII). κ_{Dio} is zero (“vanished”; Vs-s) at two different t_{RW} values: zero (where q^2 will also be zero), and at t_v , near 5 ms (Figure 3b,e). In other scenarios, the latter vanishing is quite important.² Before Vs-s, the majority (intracellular) population has the greater diffusivity. This situation switches as the system passes through the Vs-s condition. Of course, the D_{out} and ΔD t_{RW} -dependences will differ for other ρ and $\langle V \rangle$ values. The importance of this is elaborated below, and in SI.VII.

Because we set the RW p_p value, we control $\langle k_{\text{io}}(p) \rangle$ (Equation 5). Figure 3c shows the $\langle k_{\text{io}}(p) \rangle$ -dependences of displacements computed as in Figure 3a, but when k_{io} can be finite. Because any practical experiment would not be able to distinguish $\text{H}_2\text{O}_{\text{in}}$ and $\text{H}_2\text{O}_{\text{out}}$, even for $\langle k_{\text{io}}(p) \rangle = 0$, we plot the weighted average of the Figure 3a $\text{H}_2\text{O}_{\text{in}}$ and $\text{H}_2\text{O}_{\text{out}}$ curves in Figure 3c (blue). For nonzero k_{io} values, individual water molecules enter and leave cells (Figure 2), so they are inherently indistinguishable. Encouragingly, the displacements are seen to be quite $\langle k_{\text{io}}(p) \rangle$ -sensitive, and increasingly so for increasing $\langle k_{\text{io}}(p) \rangle$. As expected, the displacement curve becomes more linear as $\langle k_{\text{io}}(p) \rangle$ increases. However, it becomes truly linear (with slope $6D_0$) only when ρ and $\langle V \rangle$ approach 0 ($\text{H}_2\text{O}_{\text{free}}$). The instantaneous slope/6 values are plotted in Figure 3d,f. The D'_{WM} value increases with $\langle k_{\text{io}}(p) \rangle$. Importantly, the well-mixed condition is reached more quickly with faster exchange (as k_{io} increases). Even with non-zero k_{io} , when q becomes sufficiently large, D'_{in} can limit at k_{io}/q^2 (SI.VII). It no longer depends on any D . Of course, the Figure 3d,f $\langle k_{\text{io}}(p) \rangle$ curve family will also differ for other ρ and $\langle V \rangle$ values.

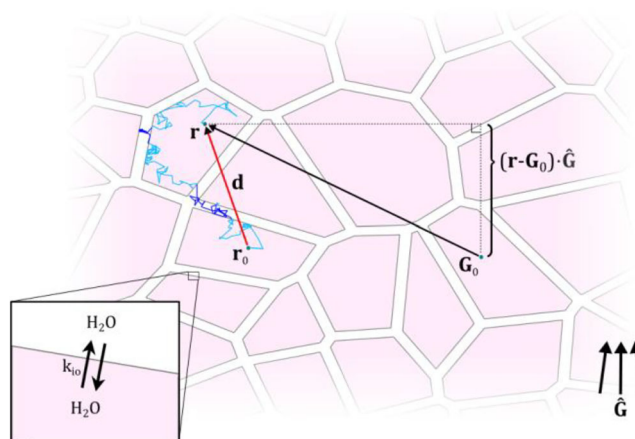


FIGURE 2 Random walk. A water molecule random walk in a Figure 1 2D slice (intracellular and extracellular steps are light and dark blue, respectively) achieves displacement d (from r_0 to r , red vector) with changed projection on a potential magnetic field gradient, $\mathbf{G}$, direction. The membrane permeation probability, p_p , is converted to k_{io} , the homeostatic cellular water efflux rate constant

We are assembling a unique, canonical, virtual library of displacements as in Figure 3c, labeled with their $\langle k_{io}(p) \rangle$, ρ , $\langle V \rangle$ parameter sets. As described next, these can be converted to D-weighted (D-w) MR decays for any MR diffusion-encoding scheme conceivable. Thus there will be a separate, unique library for each type of encoding.

For notational simplicity: hereafter $\langle k_{io}(p) \rangle \equiv k_{io}$, and $\langle V \rangle \equiv V$.

3.4 | Diffusion MR decays

For a homogeneous fluid, the D-w transverse $^1\text{H}_2\text{O}$ signal $S(b)$ decays in “b-space” (Equation 7),

$$\left(\frac{S(b)}{S_0} \right) = e^{-bD_0}, \quad (7)$$

where S_0 is $S(b=0)$ (D_0 , for pure water). The acquisition pulse sequence MR coherence dephasing potential (b) is the trace of the generalized $\mathbf{b}$ tensor: its “size”.²⁸ The generalized Stejskal–Tanner integral b expression is given in Equation 8,

$$b = \int_0^t \left[\gamma \int_0^{t'} G(t'') dt'' \right]^2 dt', \quad (8)$$

where $\int_0^{t'} G(t'') dt''$ integrates all gradients up to time t' ,⁴⁵ spanning the entire encoding time-course.^{46,47} For the (by far most common) two-PFG, single diffusion-encoding (SDE) sequence,²⁸ the integrated solution is approximated as in Equation 9,

$$b_{\text{SDE}} = b = (\gamma G \delta)^2 t_D = q^2 t_D \quad (9)$$

where $q = \gamma G \delta$, γ is the ^1H gyromagnetic ratio 2.68×10^8 rad/s/T (42.6 MHz/T), G and δ are the PFG strength and duration, respectively, the diffusion time is $t_D \equiv [\Delta - (\delta/3)]$, and Δ is the PFG pair leading edge interval. This accounts for precessional and possible diffusion dephasing during the PFGs and during the t_D diffusion period, respectively. The quantity b can be considered a linear (1D) vector.²⁸ Importantly, Equation 7 embodies a first approximation normalization for magnetic field strength (B_0) and T_2 differences. The transverse S_0 , $S(b=0)$, signal is obtained at the same t_D ($= \text{TE}$), as are the other $S(b)$ signals. Thus $S_0 = e^{-R_2 \text{TE}}$. S_0 does not represent Boltzmann magnetization. Whatever T_2 coherence dephasing obtains for different spin molecular environments has accrued by $t_D = \text{TE}$. We are not quantifying $S(b)$ or S_0 , but $S(b)/S_0$. Accordingly, the MAD1 model has no T_2 . Equation 9 formally normalizes for b -incrementation, varying either the $(\gamma G \delta)^2$ factor (constant time [CT]) or the t_D factor (constant gradient [CG]), as long as the decay is diffusion rate-limited.^{16,17,48,49} Clear literature evidence for this is shown in.⁸ The SDE-CG

acquisition can reach a b value where the decay is not diffusion-limited (SI.VIII). Although widely employed,^{28,29} most tissue anisotropic contributions are small compared with D tensor magnitude (see below).

To simulate an SDE-CT D-w MR decay, the projection of the particle starting position (r_0 , Figure 2) on $\mathbf{G}$ is noted at the beginning of each RW. The particle spin phase ($\phi = \int \omega dt_{RW}$) is advanced for $\delta \mu$ s of Larmor precession (i.e., during the first PFG) at the local field, with frequency ω determined by that projection. After $\mathbf{G}$ is switched off, the particle randomly diffuses for a period t_D (the SDE pulse sequence²⁸) to a new position, r . When the second, rephasing PFG is turned on, the spin phase is again advanced for $\delta \mu$ s of Larmor precession, but the local ω is now determined by the projection of the new r on $\mathbf{G}$. The value of $S(b)$ is determined by averaging the S values of many spins considering their displacements after undergoing RWs in different ensembles (of identical k_{io} , ρ , and V) and b_{SDE} is G -incremented (CT): δ , t_D , and Δ are 20, 43.3, and 50 ms, respectively, in our simulations. Hereafter, we designate b_{SDE} as b .

We explore our hypothesis by varying each of the three fundamental parameters in turn. Figure 4 presents families of simulated diffusion decays from 12×10^6 RWs in different ensembles with the same k_{io} , ρ , and V values and typically 160,000 cells; 400 b values are calculated out to $b_{max} = 4 \text{ ms}/(\mu\text{m})^2 (= 4000 \text{ s}/(\text{mm})^2)$,^{50,51} however, it would take averaging over many more RWs to avoid simulation artifacts manifest long before b reaches 40. MC random fluctuations can sometimes cause greatly diminished simulated signal to increase with diffusion time: for experimental decays, this would violate the second law of thermodynamics,

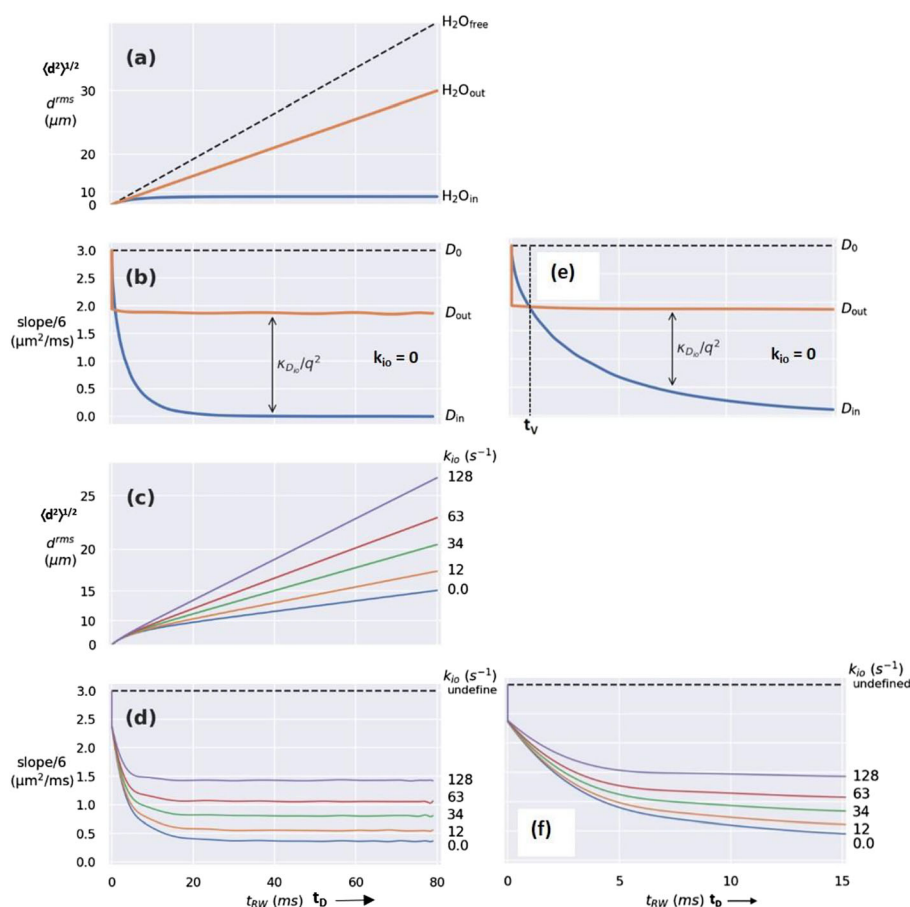


FIGURE 3 Displacements and diffusion coefficients. Simulated mean-square molecular displacements, d^{rms} , in ensembles with $\rho = 781,000$ cells/ μL and $\langle V \rangle = 1.0 \text{ pL}$. (a) When $\langle k_{io}(p) \rangle = 0 \text{ s}^{-1}$, displacements for extracellular and intracellular water (H_2O_{out} [orange] and H_2O_{in} [blue], respectively) can be distinguished. The pure water Gaussian displacement (H_2O_{free}) is also shown as the dashed line. The abscissa is the elapsed random walk time, $t_{RW} (\equiv t_D)$. (Note the squared ordinate scale.) (b) The t_{RW} -dependences of $(1/6)$ the no-exchange panel (a) curve instantaneous slopes. The curves are labeled D_{out} and D_{in} . The faster but smaller D_{out} decrease, compared with the slower but larger D_{in} decrease, is significant. Each curve reaches a limiting value; the “well-mixed” condition. $|\Delta D|$ is given as a function of the diffusigraphic shutter-speed, κ_{Dio} , and the wave number, q (when there is a diffusion-encoding gradient). (c) The t_{RW} -dependences of d^{rms} for increasing $\langle k_{io}(p) \rangle$ values. The blue $\langle k_{io}(p) \rangle = 0 \text{ s}^{-1}$ curve is the weighted average of the panel (a) curves. The displacements increase with increasing $\langle k_{io}(p) \rangle$. (d) The t_{RW} -dependences of the panel (c) curve instantaneous slopes. (e) The zoomed (b) origin region. (f) The zoomed (d) origin region. In two-site-exchange (2SX) theory, the (b)(e) D_{in} and D_{out} quantities are considered intrinsic D values (they represent the situation in the absence of exchange), while the (d)(f) D quantities are apparent D 's (because they result from exchange): these are not ADCs. Note the extremely rapid “mixing” for D_{out} (b,e). In panel (e), the t_D value at which κ_{Dio} vanishes is labeled t_v . ADC, apparent diffusion coefficient

reflecting an apparent spontaneous entropy decrease. Thus we show simulations to only $4 \text{ ms}/(\mu\text{m})^2 (= 4000 \text{ s}/(\text{mm})^2)$. Preliminary (Figure 4) versions were reported.^{1,52} The panels have ordinates that measure $\log_{10}(S(b)/S_0)$ (note the logarithmic scale), while the abscissae increment b .

For Figure 4a, ρ was fixed at $180,000 \text{ cells}/\mu\text{L}$, and V at 5.00 pL (the intracellular volume fraction, $v_i = 0.90$). Decays for five finite k_{io} values, 3 to 48 s^{-1} , are shown. The decays are nonlinear in these semi-log plots; non-Gaussian behavior. The decay for $k_{io} = \infty$ (pure water) is linear, indicating Gaussian diffusion. The negative $\{\ln(S(b)/S_0)/b\}$ slope of the latter is $D_0 = 3.0 \times 10^{-3} \text{ mm}^2/\text{s}$. Increasing k_{io} accelerates diffusivity. This is sensible: on average, water molecules can move further, and dephase more, the more permeable the cell membranes. Note the encouragingly good sensitivity to large k_{io} . For Figure 4b, k_{io} was fixed at 12.0 s^{-1} and V at 5.00 pL . Curves for five ρ values, 0 to $190,000 \text{ cells}/\mu\text{L}$, are shown. Increasing ρ decelerates diffusivity, over a large ($190,000 \text{ cells}/\mu\text{L}$) range. This is also sensible: water molecules cannot move as far, the greater the “cellularity”. The membrane “concentration”, $\langle A \rangle \rho$, increases with increasing ρ (Figure S5b). The Gaussian decay for $\rho = 0 \text{ cells}/\mu\text{L}$ (pure water) is repeated. For Figure 4c, k_{io} was fixed at 12.0 s^{-1} and ρ at $180,000 \text{ cells}/\mu\text{L}$. Decays for five V values, 0 to 5.25 pL , are shown. As generally expected, diffusivity decreases with V increase, averaged over many thousands of cells. At constant V (4b), v_i increases with ρ . At constant ρ (4c), v_i increases with V . Thus in both cases v_o decreases, giving a consequent “tortuosity” increase.

Very commonly in the literature, an arbitrary, two-point approximation of the asymptotic $\{\ln(S(b)/S_0)/b\}$ slope is made from a straight line drawn between $S(0)$ and S at $b = 1.0 \text{ ms}/(\mu\text{m})^2$ (dashed vertical lines in Figure 4). This is termed the apparent diffusion coefficient (ADC). A different choice of b yields a different ADC value.

In Figure 4, we also display quality literature⁵³ experimental data from the awake human cerebral cortex (1.5 T , 3.3 mL ROI ; points). One MADI simulation (green in each panel) matches these well. Its parameter values are $k_{io} = 12 \text{ s}^{-1}$, $\rho = 180,000 \text{ cells}/\mu\text{L}$, and $V = 5.0 \text{ pL}$. These are quite reasonable. In the companion paper,⁸ we give an extensive discussion of ex vivo cytometric measures that can potentially be used for MADI validation. It should be recalled that $k_{io} = 12 \text{ s}^{-1}$ most likely could not be measured with s-s-DCE-MRI, even if CA could cross the BBB.

4 | DISCUSSION

4.1 | Implications for literature DWI changes

The ADC is a popular estimate of the negative of the asymptotic $\{\ln(S(b)/S_0)/b\}$ slope. However, its empiricism often yields misleading or erroneous interpretations. As seen above, our ultimate goal is formal matchings of entire experimental voxel b -decays to produce k_{io} , ρ , and V parametric maps without invoking arbitrary compartmental D values.⁸ Nonetheless, we can appraise our simulations' clinical implications by considering the k_{io} -, ρ -, and V -dependences of their ADC values. We used straight lines between the $\ln(S/S_0)$ values at $b = 0$ and 1 (conventional; Figure 4, vertical dashed lines) for each Figure 4 curve. Typical results are shown in¹ and⁵² as $\text{ADC} = f(k_{io})_{\rho,V}$, $f(\rho)_{k_{io},V}$, or $f(V)_{k_{io},\rho}$ plots. The latter is, for example, the ADC V -dependence at constant k_{io} and ρ . In these, we see increasing k_{io} and (ρ and V) cause ADC increases and decreases, respectively. The ADC magnitudes are quite consistent with the reported awake human cortex and WM trD values of 0.8 and $0.7 (\mu\text{m})^2/\text{ms}$, respectively.⁵⁴ The MADI ADC parameter-dependence plots may shed light on many literature DWI observations. They immediately bring to mind ADC changes found to be acute, and those that are chronic.

We know experimental k_{io} values include the second Equation 1 term, $k_{io}(a)$.² Because this reflects a fast metabolic rate ($x^c\text{MR}_{\text{NKA}}$ can give $10^{12} \text{ H}_2\text{O}$ molecules actively cycled/ s/cell ⁸), acute ADC changes could be dominated by very fast $x^c\text{MR}_{\text{NKA}}$ changes.

4.1.1 | Stroke

Observed acute brain ADC changes include: human and cat stroke ($30\% \downarrow$, $< 2 \text{ h}$ ^{55–58}), direct rat brain ouabain (a specific NKA inhibitor) injection ($39\% \downarrow$, $< 15 \text{ min}$ ⁵⁹), sudden rat death ($42\% \downarrow$,⁶⁰ $44\% \downarrow$ ⁶¹), rat anesthesia/sedation,⁶² external sensory stimulus,⁶³ and genetically engineered, triggered specific $k_{io}(\rho)$ changes.^{64,65} Such sudden literature ADC decreases have often been interpreted in terms of V increases alone. However, the ADC k_{io} -dependence is also consistent with the rapid pH decrease⁵⁸ and slower $[\text{Na}_t^+]$ increase^{56,66} accompanying stroke as being metabolic consequences of NKA inhibition by sudden ischemic $^c\text{MR}_{\text{NKA}}$ quench. Increasing V without increasing $[\text{Na}_{in}^+]$ would cause $[\text{Na}_t^+]$ decrease,^{5,67} not increase. Although V increase does occur, it is not the cause of the stroke ADC drop.⁵⁵

4.1.2 | Cancer

By contrast, chronic ADC decreases and increases, respectively, with cancer progression²³ and conventional cytotoxic therapy,⁶⁸ are usually attributed to ρ changes, and our plots support this: the ρ value has to change by hundreds of thousands of cells/ μL (Figure 4b), and this should be slow. Beds of pure breast carcinoma and lymphocyte cells can be estimated to have ρ values of $1,250,000$ and $6,000,000 \text{ cells}/\mu\text{L}$, respectively.⁶⁹

Such estimates are made from 2D histological “cellularity”:^{8,70,71} New, targeted, cell-sparing cancer therapies may be expected to decrease k_{io} (and thus decrease tumor ADC) in days, if not hours, before there can be much ρ change.⁷²

4.2 | Implications for the molecular mechanism of water diffusion in tissue

It may be surprising to some that the use of the free water diffusion coefficient, D_0 , can lead to good matchings (Figure 4) and mappings.⁸ There have been elegant efforts to characterize water diffusion in interstitial,⁵⁵ cytosolic,^{55,73} mitochondrial,⁷³ and nuclear⁷³ spaces, with their attendant organellar membranes. The analyses often assumed k_{io} is zero, but a s-s-CA study of a suspension of tiny mitochondria ($V \approx 2$ fL) indicated that the fast-exchange-limit (Vs -s) condition obtains.⁷⁴ We estimate mitochondrial $k_{io}(p)$ at 2.4 s^{-1} , and note there is a membrane H^+ ATPase (thus $k_{io} > k_{io}(p)$). Such work has generally found that, if one had to define an ADC for these compartments, it would be smaller than D_0 , often about half.⁵⁵ If it was possible to have macroscopic samples of these compartment lumens, one would likely find Gaussian diffusion with a D less than D_0 . Studies of water in macroscopically homogeneous systems—agar and gelatin gels,⁷⁵ protein/fat/gel cheese models,⁷⁶ and cheeses themselves⁷⁷—find Gaussian diffusion with D values down to 0.8, 0.5, and 0.2 D_0 , respectively. These findings are consistent with our displacement simulations for compartmentalized spaces (Figure 3), which show instantaneous slopes varying at early diffusion time but limiting at values less than D_0 . That is, there is no well-defined overall D but, if one postulated there is, it would be less than D_0 . Choosing any single D value would be arbitrary, and would just add an empirical fourth variable. It would make our approach less parsimonious. Although the instantaneous D_{in} and D_{out} would start at a smaller value in Figure 3b,e, they would still fall to the same limiting values, determined by V and ρ for D_{out} . Our model returns D_0 for appropriate tissues ($\rho = V = 0$, $k_{io} = \infty$; fig. 2 in⁸).

This can be further appraised with the Monte Carlo Equation 5. We rewrite it as Equation 10.

$$\frac{\langle k_{io}(p) \rangle}{\langle A/V \rangle} = p_p \left(\frac{D}{6t_s} \right)^{1/2} = (p_p^2 D)^{1/2} (6t_s)^{-1/2} = p_p \left(\frac{D}{l_s} \right) = (p_p D) (l_s)^{-1} \quad (10)$$

For any particular d^{rms} time-course simulation, the ratio of the mean $\langle k_{io}(p) \rangle$ to the mean claustrophobic ratio $\langle A/V \rangle$ is constant (Figure 3). Thus the factors $(p_p^2 D)^{1/2}$ and $(p_p D)$ are constant, for given t_s and l_s values, respectively, which are the same on each side of the membrane. If we arbitrarily choose a D value different from D_0 during data matching, it will be exactly compensated by p_p . A $(D_0/(6t_s))^{1/2}$ decrease from 22.4 mm/s would be compensated by a p_p increase from its typical value of 0.005. A 50% D reduction is a 29% $D^{1/2}$ reduction. The MADI parameter values would be unaffected. The real test lies in independent V , ρ , and k_{io} validation, and MADI seems to do rather well in this regard.⁸ Because we want D to approach D_0 as ρ approaches 0 (pure water; fig. 2e, MADI II.⁸), we choose $D = D_n = D_0$.

It is true that uncompartimented cytosol models exhibit Gaussian D_n values less than D_0 . However, for a compartmentalized sample, using these would affect mostly the intracompartimental mixing time, more than the limiting intrinsic D_{WM} value (Figure 3): but the latter dominates common diffusion periods, and we note these simulations show why the pertinent limiting D values in compartmented tissue are less than D_0 . For these reasons, we also cannot set different nanoscopic D values, or (D/l_s) values, for inside and outside water using Monte Carlo techniques. For steady-state exchange, p_p must be the same on each side of the membrane. Also, if we could reduce the D value, we would have an additional,

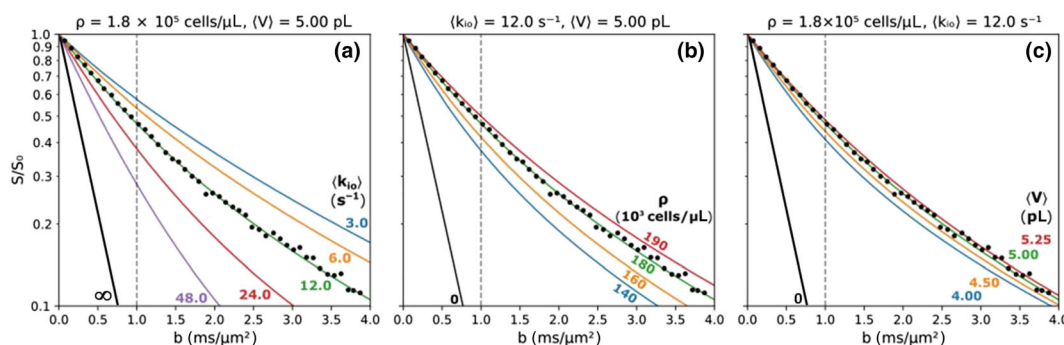


FIGURE 4 Simulated MR diffusion decays. Semi-log plots of simulated SDE-CT diffusion-weighted b -space decays. Each metabolic activity diffusion imaging (MADI) model parameter (k_{io} , ρ , $\langle V \rangle$) is varied in turn, with the other two fixed (given above each panel). Increasing k_{io} , ρ , and $\langle V \rangle$, respectively, causes diffusivity to (a) increase, (b) decrease, and (c) decrease. The linear pure water Gaussian decay (k_{io} undefined, $\rho = 0$, $\langle V \rangle = 0$) is plotted in each panel. Literature⁵³ experimental data from the awake, resting human cerebral cortex (1.5 T, 3.3 mL ROI; points) are also plotted. One MADI-simulated decay (green in each panel) matches these well. Its parameter values are $k_{io} = 12 \text{ s}^{-1}$, $\rho = 180,000 \text{ cells}/\mu\text{L}$, and $\langle V \rangle = 5.0 \text{ pL}$. The vertical dashed lines mark the b value, $1 \text{ ms}/(\mu\text{m})^2 (= 1000 \text{ s}/(\text{mm})^2)$, conventionally (but arbitrarily) used to evaluate the ADC. ADC, apparent diffusion coefficient; ROI, region of interest

empirical fitting parameter, and two if we assumed different D_{in} and D_{out} values. Figures S6 and S7 elaborate different ways of $\langle A/V \rangle$ averaging in the Voronoi ensembles.

That D values are t_D -dependent in a compartmentalized sample has been realized for a long time.^{43,44} The blue (D_{in}) curve in Figure 3b,e is a manifestation of the non-Gaussian diffusion dispersion, or “diffusion spectrum”.^{47,78} D is time-independent for only Gaussian diffusion. The nanoscopic D (D_n) values are mainly pertinent only very fleetingly (Figure 3a,c), during what we refer to as “mixing” periods. These are the times during which the intrinsic D_{in} and D_{out} curves drop to limiting values (Figure 3b,e): the latter reflect the “well-mixed” (or steady-state) NMR condition.⁷ This can be understood as the situation when most molecules have encountered membranes many times. Our simulation is consistent with recent experimental brain white matter⁷⁹ diffusion dispersion time-courses.

It is important to note the orange intrinsic D_{out} curve in Figure 3b,e drops to its limiting value essentially instantaneously (within ns). Figure 2 (dark blue trajectories) illustrates this is because the extracellular A/V (claustrophobia) ratio is large. We will see this is likely also the case for intracellular water in all regions of dendritic neural cells,⁸ and has very fortunate consequences for brain MADI. The Figure 3 blue intrinsic D_{in} curve requires ~ 20 ms, and this is understood via Equation 6. The equation shows it is mainly this period that depends inversely on D_n (D_0 here), and increases with $\langle V \rangle$ (more generally, $\langle V/A \rangle$). Thus, for most tissues, it is intracellular diffusion that dominates the diffusion dispersion time-period, consistent with a longer mixing period for larger skeletal muscle cells.⁸⁰ For most of the diffusion period (essentially all of it for extracellular water), it is the limiting intrinsic D_{WM} 's that are relevant, and these have little to do with D_0 . Once the well-mixed condition is achieved, membrane permeation becomes diffusion-limiting. This is optimal for our purposes. The D_{WM} values distinguish intracellular and extracellular water more than the time-dependent D_n values do. For D_{out} , D_{WM} is nonzero. For D_{in} , $D_{WM} = 0$. As stated above, the limiting intrinsic D_{in} must be zero and the limiting intrinsic D_{out} value depends on the ensemble ρ and V values.

It is important to note Figure 3d,f show that the mixing periods are only shortened by the presence of finite trans-cytolemmal water exchange. Once diffusion has entered the homeostatic, permeation-limited regime, even with nonzero k_{io} , D'_{in} can limit at k_{io}/q^2 when q becomes sufficiently large (SI.VII). D'_{in} no longer depends on any diffusion coefficient. D'_{out} is a more complicated function of k_{io} , V , and ρ .

In Equation 1, the $k_{io}(\rho)$ and $k_{io}(a)$ terms have important $\langle A/V \rangle$ and $\langle 1/V \rangle$ factors, respectively. These approach $+\infty$ as $V \rightarrow 0$ (the very similar V -dependences of $\langle A/V \rangle$ and $\langle 1/V \rangle$ are shown in Figure S6b). This is why k_{io} is undefined for pure water, and approaches $+\infty$.

Our experience indicates that two variables are generally insufficient to match diffusion decays.⁸ However, there are many successful matchings²³ with the empirical, bi-exponential Equation 11,

$$S/S_0 = p'_A e^{-q^2 t_D D'_A} + p'_B e^{-q^2 t_D D'_B} \quad (11)$$

where p'_A , p'_B , D'_A , and D'_B are the apparent mole fractions (“populations”) and diffusion coefficients, respectively, of water molecules in compartments A and B. Because $p'_B \equiv 1 - p'_A$, there are three variables. Because these (p'_A , D'_A , and D'_B) are empirical, no molecular interpretation can be verified: consequently, any attempt is fraught with peril.^{17,81} It is important to note Equation 11 inherently assumes no intercompartmental molecular exchange (i.e., $k_{AB} = 0$). The Kärger/Andrasko [KA] model (reviewed in¹⁷) was introduced to add only one additional (exchange) parameter, while yielding an effectively infinite number of exponentials. Our Figure 3 results, however, reveal significant underlying challenges for KA (SI.VII).

Thoughtful observations in the DWI literature lament the field lacks a “clear understanding of the mechanisms responsible”²³ and that “general discrepancies between models...and DWI experiments...are problematic”.⁸² In our results, the probabilities of encountering (ρ , V) and permeating (k_{io}) cell membranes seem sufficient to describe tissue water diffusion well: no empirical trans-membrane “viscosity” disparities (different D s) are necessary. Consistent with this, water molecules near and in the larger neuronal cell body (soma) ($\langle V \rangle$, ~ 1 pL⁸³) exhibit greater diffusivity than water molecules in surrounding, cell membrane-rich, highly dendritic neuropil regions.⁸⁴ (Possible implications of this for brain ρ and V values are given in.⁸) Membrane “concentration” and permeability⁸⁵ appear to be diffusion-limiting. In Figure S5b, we show increasing ρ causes increasing a_e , the ensemble surface area “concentration”. The quantity a_e is defined as $\langle A \rangle \rho$, and has units of μm^{-1} : it is analogous to $\langle V \rangle \rho = v_i$. In general, the claustrophobia ratio $\langle A/V \rangle$ (Equation 5) decreases as V increases (Figure S6b), but for a given V , the spherical shape has the smallest $\langle A/V \rangle$. Because $\langle A/V \rangle$ is also a measure of membrane concentration, this suggests encountering a cell membrane is the diffusion-limiting step for water in tissue. A chemical kinetics theorem for serial processes states all steps preceding the rate-limiting one appear effectively infinitely fast. Although diffusion of water in tissue compartmental media is surely slower than in pure water, it still seems fast compared with encountering and permeating cytolemmal membranes. It might as well be modeled as having the pure water diffusion coefficient, D_0 . In general, after a certain period, the well-mixed condition obtains for intracellular and extracellular water.

In the companion paper, we show further data matchings for a broad array of normal and pathological tissues.⁸ These, and the Figure 4 matching, suggest the forward problem for our MADI model—can it describe tissue water SDE-DWI diffusion decays—is well in hand. There is also an extensive discussion of results from literature ex vivo cytometric methods that have the potential to validate MADI parameter accuracy.⁸

Of course, the inverse problem—the best-matching parameter set uniqueness for a given experimental decay from a discrete library (i.e., precision)—is more difficult. Thus, also in,⁸ we show initial proof-of-concept, noninvasive, high-resolution MADI k_{io} , ρ , and V mappings of the

awake, resting normal human brain. The $k_{\text{iso}} \bullet V$ product map correlates, both quantitatively and qualitatively, with the pharmacokinetic ^{18}F FDG-PET map measuring homeostatic brain glucose consumption.⁸

ACKNOWLEDGMENTS

We thank Professors Valerie Anderson, Deborah Burstein, Alexander Guimaraes, Wei Huang, Christopher Kroenke, Philip Kuchel, Charles Landis, Jing-Huei Lee, Manoj Sammi, Rosalie Sears, Janio Sklaruk, Mark Woods, Cory Wyatt, Drs. Dana Andersen, Joseph Armstrong, John Grinstead, Randall Jones, Maren Laughlin, and Ethan Ulrich, and Messrs. Eric Baetscher and Arvin Paranjpe, for stimulating discussions.

FUNDING

OHSU Brenden-Colson Center for Pancreatic Care, Sheppard (PI); Springer (PI); Li (PI). [CSS, XL, EMB, BM].

NIH UL1TR02369, Ellison, Morris (PIs). Awarded by the Oregon Clinical and Translation Institute, Biomedical Innovation Program, Li (PI); Pike (PI). [XL, CSS, EMB, BM].

Bracco Diagnostics, Inc., JHM (PI). [JHM, GJW].

CONFLICT OF INTEREST

CSS, EMB, XL, BM, GJW, TMB, WDR, and JHM are inventors on U.S. Provisional Patent Application (No. U.S. 62/482,520) "Activity MRI."

DATA AVAILABILITY STATEMENT

All data are available in the main text or in the supplementary material.

ORCID

Charles S. Springer Jr  <https://orcid.org/0000-0002-5966-2135>

Xin Li  <https://orcid.org/0000-0002-9283-3602>

REFERENCES

- Springer CS. Using $^1\text{H}_2\text{O}$ to measure and map sodium pump activity *in vivo*. *J Magn Reson*. 2018;291:110-126. doi:10.1016/j.jmr.2018.02.018
- Li X, Mangia S, Lee J-H, Bai R, Springer CS. NMR shutter-speed elucidates apparent population inversion of $^1\text{H}_2\text{O}$ signals due to active transmembrane water cycling. *Magn Reson Med*. 2019;82:411-424. doi:10.1002/mrm.27725
- Bai R, Springer CS, Plenz D, Basser PJ. Brain active trans-membrane water cycling measured by MR is associated with neuronal activity. *Magn Reson Med*. 2019;81:1280-1295. doi:10.1002/mrm.27473
- Bai R, Springer CS, Plenz D, BPJ. Fast Na^+/K^+ pump driven, steady-state transcytolemmal water exchange found in neuronal tissue: A study of rat brain cortical cultures. *Magn Reson Med*. 2018;79:3207-3217. doi:10.1002/mrm.26980
- Rooney WD, Li X, Sammi MK, Bourdette DN, Neuwelt EA, Springer CS. Mapping human brain capillary water lifetime: High-resolution metabolic neuroimaging. *NMR Biomed*. 2015;28:607-623. doi:10.1002/nbm.3294
- Springer CS, Li X, Tudorica LA, et al. Intratumor mapping of intracellular water lifetime: Metabolic images of breast cancer? *NMR Biomed*. 2014;27:760-773. doi:10.1002/nbm.3111
- Zhang Y, Poirier-Quinot M, Springer CS, Balschi JA. Active trans-plasma membrane water cycling in yeast is revealed by NMR. *Biophys J*. 2011;101:2833-2842. doi:10.1016/j.bpj.2011.10.035
- Springer CS Jr, Baker EM, Li X, et al. Metabolic activity diffusion imaging [MADI]: II. Non-invasive, high-resolution human brain imaging of sodium pump flux and cell metrics. *NMR Biomed*. 2022;e4782. doi:10.1002/nbm.4782
- Vinakota KC, Bassingthwaite JB. Myocardial density and composition: A basis for calculating intracellular metabolite concentrations. *Am J Physiol Heart Circ Physiol*. 2004;286:H1742-H1749.
- Landis CS, Li X, Telang FW, et al. Equilibrium transcytolemmal water exchange kinetics in skeletal muscle *in vivo*. *Magn Reson Med*. 1999;42:467-478.
- Schmidt K, Bau M, Merschel G, Tepe N. Anthropogenic gadolinium in tap water-based beverages from fast-food franchises in six major cities in Germany. *Sci Total Environ*. 2019;687:1401-1408. doi:10.1016/j.scitotenv.2019.07.075
- Kiviniemi A, Gardberg M, Ek P, Fratzén J, Bobacka J, Minn H. Gadolinium retention in gliomas and adjacent normal brain tissue: Association with tumor contrast enhancement and linear/macrocyclic agents. *Neuroradiology*. 2019;61:535-544. doi:10.1007/s00234-019-02172-6
- Li LZ, Zhou R, Xu HN, et al. Quantitative magnetic resonance and optical imaging biomarkers of melanoma metastatic potential. *Proc Natl Acad Sci*. 2009;106:6608-6613. doi:10.1073/pnas.0901807106
- Ruggiero MR, Baroni S, Pezzana S, Ferrante G, Crich SG, Aime S. Evidence for the role of intracellular water lifetime as a tumor biomarker obtained by *in vivo* field-cycling relaxometry. *Angew Chem Int Ed Engl*. 2018;57:7468-7472. doi:10.1002/anie.201713318
- Ruggiero MR, Baroni S, Aime S, Crich SG. Relaxometric investigations addressing the determination of intracellular water lifetime: A novel tumor biomarker of general applicability. *Mol Phys*. 2019;117:968-974. doi:10.1080/00268976.2018.1527045
- Pfeuffer J, Flögel U, Dreher W, Leibfritz D. Restricted diffusion and exchange of intracellular water: Theoretical modelling and diffusion time dependence of ^1H NMR measurements on perfused glial cells. *NMR Biomed*. 1998;11:19-31. doi:10.1002/(SICI)1099-1492(199802)11:13.O.CO;2-O
- Lee J-H, Springer CS. Effects of equilibrium exchange on diffusion-weighted NMR signals: The diffusigraphic 'shutter-speed'. *Magn Reson Med*. 2003;49:450-458.
- Huang W, Moloney B, Li X, et al. Active trans-membrane water cycling reflects metabolic activity in breast cancer. *Proc Int Soc Magn Reson Med*. 2020;28:4798.

19. Li X, Priest RA, Woodward WJ, et al. Cell membrane water exchange effects in prostate DCE-MRI. *J Magn Reson*. 2012;218:77-85.
20. Landis CS, Li X, Telang FW, et al. Determination of the MRI contrast agent concentration time course *in vivo* following bolus injection: Effect of equilibrium transcytolemmal water exchange. *Magn Reson Med*. 2000;44:563-574.
21. Buckley DL. Shutter-speed dynamic contrast-enhanced MRI: Is it fit for purpose? *Magn Reson Med*. 2019;81:976-988. doi:10.1002/mrm.27456
22. Li X, Rooney WD, Springer CS. A unified magnetic resonance imaging pharmacokinetic theory: Intravascular and extracellular contrast reagents. *Magn Reson Med*. 2005;54:1351-1359. [Erratum: *Magn Reson Med*. 2006;55:1217.]. doi:10.1002/mrm.20684
23. Iima I, Le Bihan D. Clinical intravoxel incoherent motion and diffusion MR imaging: Past, present, and future. *Radiology*. 2016;278:13-32. doi:10.1148/radiol.2015150244
24. van der Walt S, Colbert SC, Varoquaux G. The NumPy array: A structure for efficient numerical computation. *Comput Sci Eng*. 2011;13:22-30.
25. Virtanen P, Gommers R, Oliphant TE, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nat Methods*. 2020;17:261-272. doi:10.1038/s41592-019-0686-2
26. Baker EM, Moloney B, Li X, Gilbert EW, Springer CS. Random walks in stochastic geometries. *Proc Int Soc Magn Reson Med*. 2019;27:3612.
27. Ahlgren A, Knutsson L, Wirestam R, et al. Quantification of microcirculatory parameters by joint analysis of flow-compensated and non-flow-compensated intravoxel incoherent motion (IVIM) data. *NMR Biomed*. 2016;29:640-649. doi:10.1002/nbm.3505
28. Topgaard D. Multidimensional diffusion MRI. *J Magn Reson*. 2017;275:98-113. doi:10.1016/j.jmr.2016.12.007
29. Verratt J, Nunes D, Rudrapatna U, et al. Noninvasive quantification of axon radii using diffusion MRI. *Elife*. 2020;9:e49855. doi:10.7554/eLife.49855
30. Wilson GJ, Springer CS, Maki JH. Translocation of water molecules in tissue. *Proc Int Soc Magn Reson Med*. 2017;25:1738.
31. Brusini L, Menegaz G, Nilsson M. Monte Carlo simulations of water exchange through myelin wraps: Implications for diffusion MRI. *IEEE Trans Med Imag*. 2019;38:1438-1445. doi:10.1109/TMI.2019.2894398
32. Rensonnet G, Scherrer B, Girard G, et al. Towards microstructure fingerprinting: Estimation of tissue properties from a dictionary of Monte Carlo diffusion MRI simulations. *Neuroimage*. 2019;184:964-980. doi:10.1016/j.neuroimage.2018.09.076
33. Winters KV, Reynaud O, Novikov DS, Fieremans E, Kim SG. Quantifying myofiber integrity using diffusion MRI and random permeable barrier modeling in skeletal muscle growth and Duchenne muscular dystrophy model in mice. *Magn Reson Med*. 2018;80:2094-2108. doi:10.1002/mrm.27188
34. Tian X, Li H, Jiang X, Xie J, Gore JC, Xu J. Evaluation and comparison of diffusion MR methods for measuring apparent transcytolemmal water exchange rate constant. *J Magn Reson*. 2017;275:29-37. doi:10.1016/j.jmr.2016.11.018
35. Li H, Jiang X, Xie J, Gore JC, Xu J. Impact of transcytolemmal water exchange on estimates of tissue microstructural properties derived from diffusion MRI. *Magn Reson Med*. 2017;77:2239-2249. doi:10.1002/mrm.26309
36. Li J-R, Nguyen HT, Van Nguyen D, Haddar H, Coatléven J, Le Bihan D. Numerical study of a macroscopic finite pulse model of diffusion MRI signal. *J Magn Reson*. 2014;248:54-65.
37. Van Nguyen D, Li J-R, Grebenkov D, Le Bihan D. A finite elements method to solve the Bloch-Torrey equation applied to diffusion magnetic resonance imaging. *J Comput Phys*. 2014;263:283-302.
38. Novikov DS, Fieremans E, Jensen JH, Helpert JA. Random walks with barriers. *Nat Phys*. 2011;7:508-514. doi:10.1038/nphys1936
39. Fieremans E, Novikov DS, Jensen JH, Helpert JA. Monte Carlo study of a two-compartment exchange model of diffusion. *NMR Biomed*. 2010;23:711-724. doi:10.1002/nbm.1577
40. Helpert JA, Ordidge RJ, Knight RA. The effect of cell membrane water permeability on the apparent diffusion coefficient of water. *Proc Soc Magn Med*. 1992;11:1201.
41. Regan DG, Kuchel PW. Mean residence time of molecules diffusing in a cell bounded by a semi-permeable membrane: Monte Carlo simulations and an expression relating membrane transition probability to permeability. *Eur Biophys J*. 2000;29:221-227.
42. Chen S-T, Springer CS. Ionophore-catalyzed cation transport between phospholipid inverted micelles manifest in DNMR. *Biophys Chem*. 1981;14:375-388.
43. Le Bihan D, Turner R, Moonen CTW, Pekar J. Imaging of diffusion and microcirculation with gradient sensitization: Design, strategy, and significance. *J Magn Reson Imaging*. 1991;1:7-28.
44. Neuman CH. Spin echo of spins diffusing in a bounded medium. *J Chem Phys*. 1974;80:4508-4511. doi:10.1063/1.1680931
45. Solomon JE, Shemesh N, Frydman L. Diffusion weighted MRI by spatiotemporal encoding: Analytical description and *in vivo* validations. *J Magn Reson*. 2013;232:76-86. doi:10.1016/j.jmr.2013.02.014
46. Does MD, Parsons EC, Gore JC. Oscillating gradient measurements of water diffusion in normal and globally ischemic rat brain. *Magn Reson Med*. 2003;49:206-215.
47. Gore JC, Xu J, Colvin DC, Yankeelov TE, Parsons EC, Does MD. Characterization of tissue structure at varying length scales using temporal diffusion spectroscopy. *NMR Biomed*. 2010;23:745-756. doi:10.1002/nbm.1531
48. Hills BP, Godward J, Manning CE, Biechlin JL, Wright KM. Microstructural characterization of starch systems by NMR relaxation and Q-space microscopy. *Magn Reson Imaging*. 1998;16:557-564.
49. Meier C, Dreher W, Leibfritz D. Diffusion in compartmental systems. I. A comparison of an analytical model with simulations. *Magn Reson Med*. 2003;50:500-509.
50. McNab JA, Edlow BL, Witzel T, et al. The human connectome project and beyond: Initial applications of 300 mT/m gradients. *Neuroimage*. 2013;80:234-245. doi:10.1016/j.neuroimage.2013.05.074
51. Huang SY, Tobyn SM, Nummenmaa A, et al. Characterization of axonal disease in patients with multiple sclerosis using high-gradient-diffusion MR imaging. *Radiology*. 2016;280:244-251. doi:10.1148/radiol.2016151582
52. Moloney B, Baker EM, Li X, Gilbert EW, Springer CS. Fundamental cell biology properties underlying *in vivo* DWI. *Proc Int Soc Magn Reson Med*. 2020;28:4506.
53. Clark CA, Le Bihan D. Water diffusion compartmentation and anisotropy at high b values in the human brain. *Magn Reson Med*. 2000;44:852-859.
54. Pierpaoli C, Jezzard P, Basser PJ, Barnett S, Di Chiro G. Diffusion tensor MR imaging of the human brain. *Radiology*. 1996;201:637-648. doi:10.1148/radiology.201.3.8939209
55. Ackerman JJH, Neil JJ. The use of MR-detectable reporter molecules and ions to evaluate diffusion in normal and ischemic brain. *NMR Biomed*. 2010;23:725-733. doi:10.1002/nbm.1530
56. Hussein MS, Stobbe RW, Bhagat YA, et al. Sodium imaging intensity increases with time after human ischemic stroke. *Ann Neurol*. 2009;66:55-62.

57. Davis D, Ulatowski J, Eleff S, et al. Rapid monitoring of changes in water diffusion coefficients during reversible ischemia in cat and rat brain. *Magn Reson Med*. 1994;31:454-460.
58. Decanniere C, Eleff S, Davis D, van Zijl PM. Correlation of rapid changes in the average water diffusion constant and the concentrations of lactate and ATP breakdown products during global ischemia in cat brain. *Magn Reson Med*. 1995;34:343-352.
59. Veldhuis WB, van der Stelt M, Delmas F, et al. *In vivo* excitotoxicity induced by ouabain, a Na^+/K^+ -ATPase inhibitor. *J Cereb Blood Flow Metab*. 2003;23:62-74.
60. Meier C, Dreher W, Leibfritz D. Diffusion in compartmental systems. II. Diffusion-weighted measurements of rat brain tissue *in vivo* and postmortem at very large b -values. *Magn Reson Med*. 2003;50:510-514.
61. Pfeuffer J, Provencher SW, Gruetter G. Water diffusion in rat brain *in vivo* as detected at very large b values is multicompartmental. *MAGMA*. 1999;8:98-108.
62. Abe Y, Tsurukazu T, Le Bihan D. Water diffusion closely reveals neural activity status in rat brain *loci* affected by anesthesia. *PLOS Biol*. 2017;15:e2001494. doi:10.1371/journal.pbio.2001494
63. Le Bihan D, Urayama S, Aso T, Hanakawa T, Fukuyama H. Direct and fast detection of neuronal activation in the human brain with diffusion MRI. *Proc Natl Acad Sci*. 2006;103:8263-8268.
64. Schilling F, Ros S, Hu D-E, et al. MRI measurements of reporter-mediated increases in transmembrane water exchange enable detection of a gene reporter. *Nature Biotech*. 2017;35:75-80. doi:10.1038/nbt.3714
65. Mukherjee A, Wu D, Davis HC, Shapiro MG. Non-invasive imaging using reporter genes altering cellular water permeability. *Nature Comm*. 2016;7:13891. doi:10.1038/ncomms13891
66. Neumaier-Probst E, Konstandin S, Ssozi J, et al. A double-tuned $^1\text{H}/^{23}\text{Na}$ resonator allows ^1H -guided ^{23}Na -MRI in ischemic stroke patients in one session. *Int J Stroke*. 2015;10:56-61. doi:10.1111/ijis.12547
67. Burstein D, Springer CS. Sodium MRI revisited. *Magn Reson Med*. 2019;82:521-524.
68. Moffat BA, Chenevert TL, Lawrence TS, et al. Functional diffusion map: A noninvasive MRI biomarker for early stratification of clinical brain tumor response. *Proc Natl Acad Sci*. 2005;102:5524-5599.
69. Yuan Y, Failmezger H, Rueda OM, et al. Quantitative image analysis of cellular heterogeneity in breast tumors complements genomic profiling. *Sci Transl Med*. 2012;4:157ra143. doi:10.1126/scitranslmed.3004330
70. Jiang X, Li H, Xie J, et al. *In vivo* imaging of cancer cell size and cellularity using temporal diffusion spectroscopy. *Magn Reson Med*. 2017;78:156-164.
71. Waterton JC, McIntyre D, Blé F-X, et al. Does ADC better predict tumor cellularity or necrosis? A multi-centre multi-vendor study in twelve rodent tumour models. *Proc Int Soc Magn Reson Med*. 2019;27:509.
72. Springer CS, Li X, Jayatilake ML, et al. Metabolic imaging of early tumor therapy. *Proc Int Soc Magn Reson Med*. 2015;23:3860.
73. García-Pérez AI, López-Beltrán EA, Kluner P, Luque J, Ballesteros P, Cerdán S. Molecular crowding and viscosity as determinants of translational diffusion of metabolites in subcellular organelles. *Arch Biochem Biophys*. 1999;362:329-338.
74. Decking UKM, Pai VM, Wen H, Balaban RS. Does binding of Gd-DTPA to myocardial tissue contribute to late enhancement in a model of acute myocardial infarction? *Magn Reson Med*. 2003;49:168-171.
75. Woessner DE, Snowden BS, Chiu Y-C. Pulsed NMR study of the temperature hysteresis in the agar-water system. *J Colloid Interface Sci*. 1983;93:283-289.
76. Métails A, Cambert M, Riaubanc A, Mariette F. Effects of casein and fat content on water self-diffusion coefficients in casein systems: A pulsed field gradient nuclear magnetic resonance study. *J Agric Food Chem*. 2004;52:3988-3995.
77. Callaghan PT, Jolley KW, Humphrey RS. Diffusion of fat and water in cheese as studied by pulsed field gradient nuclear magnetic resonance. *J Colloid Interface Sci*. 1983;93:521-529.
78. Jiang X, Li H, Xie J, Zhao P, Gore JC, Xu J. Quantification of cell size using temporal diffusion spectroscopy. *Magn Reson Med*. 2016;75:1076-1085.
79. Fieremans E, Burcaw LM, Lee H-H, Lemberskiy G, Veraart J, Novikov DS. *In vivo* observation and biophysical interpretation of time-dependent diffusion in human white matter. *Neuroimage*. 2016;129:414-427. doi:10.1016/j.neuroimage.2016.01.018
80. Lemberskiy G, Feiweier G, Gyftopoulos S, Axel L, Novikov DS, Fieremans E. Assessment of myofiber microstructure changes due to atrophy and recovery with time-dependent diffusion MRI. *NMR Biomed*. 2021;34:e4534. doi:10.1002/nbm.4534
81. Mulkern RV, Balasubramanian M, Maier SE. On the perils of multiexponential fitting of diffusion MR data. *J Magn Reson Imaging*. 2017;45:1545-1547.
82. Nevo U, Özasan E, Komlos ME, Koay CG, Sarlls JE, Bassar PJ. A system and mathematical framework to model shear flow effects in biomedical DW-imaging and spectroscopy. *NMR Biomed*. 2010;23:734-744. doi:10.1002/nbm.1591
83. Oberheim NA, Takano T, Han X, et al. Uniquely hominid features of adult human astrocytes. *J Neurosci*. 2009;29:3276-3287.
84. Flint JJ, Menon K, Hansen B, Forder J, Blackband SJ. Visualization of live, mammalian neurons during Kainate-infusion using magnetic resonance microscopy. *Neuroimage*. 2020;219:116997. doi:10.1016/j.neuroimage.2020.116997
85. Williamson NH, Ravin R, Benjamini D, et al. Magnetic resonance measurements of sub-cellular membrane structures in live and fixed neural tissue. *eLife*. 2019;8:e51101. doi:10.1101/694661

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Springer CS Jr, Baker EM, Li X, et al. Metabolic activity diffusion imaging (MADI): I. Metabolic, cytometric modeling and simulations. *NMR in Biomedicine*. 2023;36(1):e4781. doi:10.1002/nbm.4781